

CivicMitra

Complete Requirements Analysis

Comprehensive Product & Technical Specification

Target: Smart India Hackathon (SIH)

Focus: Jharkhand Pilot with Multi-State Extensibility

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1. Executive Summary

CivicMitra is a comprehensive civic issue management platform designed to bridge the gap between citizens and municipal authorities. The system enables citizens to report geo-tagged civic issues through photo and voice submissions via a mobile-first interface, while providing municipal staff with powerful tools to receive, route, act on, and monitor these reports with transparent SLAs and comprehensive analytics.

Key Benefits

- **Citizen Empowerment:** Easy, accessible reporting mechanism with real-time tracking
- **Municipal Efficiency:** Automated routing, SLA management, and comprehensive analytics
- **Transparency:** Public tracking and audit trails for accountability
- **Scalability:** Designed for multi-state deployment with local customization

2. Project Goal

Mission Statement: Enable citizens to report geo-tagged civic issues (photo/voice) via a mobile-first interface and let municipal staff receive, route, act on, and monitor those reports with transparent SLAs and analytics.

3. High-Level Functional Areas

- **Report Submission:** Mobile-first PWA + web fallback
- **Public Tracking:** By tracking ID / phone
- **Admin Dashboard:** List, map, detail, assignment
- **Auto-routing Engine:** Category + ward lookup
- **Field Officer Mobile App:** Task list, upload closure proof
- **Notifications:** SMS / push / email
- **Analytics & Reporting:** Heatmap, KPIs
- **Security & Audit:** RBAC, immutable history

4. Functional Requirements

MVP (Must-Have)

FR-1: Citizen Report Creation

Citizen can create a new report with: photo (1), category, short text, auto-GPS (or manual pin), optional phone/email, anonymous toggle.

FR-2: Tracking ID Generation

System returns a tracking ID and shows a confirmation screen.

FR-3: Public Report Tracking

Citizen can track a report status via tracking ID + phone.

FR-4: Admin Dashboard List View

Admin dashboard: list view of reports with filters (status, category, ward, date).

FR-5: Report Detail Management

Report detail view: photo, map, metadata, history, assign button, status change (New → Acknowledged → Assigned → In Progress → Resolved).

FR-6: Auto-routing System

Simple rule-based routing: category → department + ward lookup.

FR-7: Field Officer Task Management

Field officer receives assigned tasks and can update status + upload closure photo.

FR-8: SMS Notifications

Notifications via SMS (mockable) to reporter at key status changes.

FR-9: Basic Analytics

Basic analytics page showing counts by category, avg resolution time, open vs closed, and a simple heatmap image (mock) per ward.

FR-10: Audit Trail

Audit trail: every status/assignment change is stored with timestamp and actor.

Important (Post-MVP)

FR-11: Enhanced Media Support

Multiple image uploads + voice note upload.

FR-12: Duplicate Detection

Duplicate detection (geospatial + time + image similarity hint).

FR-13: Offline Capability

Offline-capable PWA with background sync.

FR-14: Advanced Routing

Advanced routing with SLA, escalation & auto-escalation if SLA misses.

FR-15: Report Export

Exportable CSV/Excel reports and scheduled exports.

FR-16: Admin Configuration

Admin configurable categories, SLA settings, department mapping.

FR-17: Citizen Feedback

Citizen feedback/rating on closed reports.

Nice-to-Have / Future**FR-18: ML Integration**

ML-based priority scoring and image-assisted classification.

FR-19: System Integration

Integration with existing municipal CRMs / asset registries.

FR-20: Multi-channel Intake

Multi-channel intake (WhatsApp bot integration, IVR).

5. Non-Functional Requirements (NFRs)

NFR ID	Category	Requirement
NFR-1	Availability	99.9% for public APIs
NFR-2	Performance	Report submission <3s on 4G for 1MB compressed photo
NFR-3	Scalability	Handle 100k reports/month and burst of 10k concurrent users
NFR-4	Security	TLS1.3, AES-256 at rest, RBAC, 2FA for admins
NFR-5	Privacy & Compliance	Data residency in India, PII minimized, opt-in for contact
NFR-6	Reliability	Durable audit logs (append-only, exportable)
NFR-7	Accessibility	WCAG 2.1 AA, multilingual (Hindi + English + local)
NFR-8	Maintainability	Infrastructure as code, CI/CD, tests
NFR-9	Observability	Metrics, logs, error tracing, dashboards (Prometheus/Grafana or equivalent)

6. User Stories & Acceptance Criteria

US-1: Citizen Submit Report

As a citizen, I want to submit a photo-backed report with my location **so that** the municipality can fix the issue.

Acceptance Criteria:

- Form accepts photo (jpg/png), category, description; auto-detects GPS and allows pin-edit
- On submit, system returns tracking ID and sends confirmation SMS (or shows onscreen)
- Report appears in admin dashboard within 5s (in demo mode)

US-2: Anonymous Submission

As a citizen, I want to submit anonymously **so that** my identity is not exposed.

Acceptance Criteria:

- Optional reporter fields (name, phone, email)
- Anonymous toggle hides personal info in admin UI

US-3: Public Tracking

As a citizen, I want to check my report status using tracking ID **so that** I can know progress.

Acceptance Criteria:

- Enter tracking ID (and phone if private) shows latest status and history

US-4: Admin View & Assign

As an admin, I want to view, filter, and assign reports **so that** my team can act.

Acceptance Criteria:

- Admin can filter by status/category/ward/date and assign to a user or team
- Assignment triggers notification to field officer

US-5: Field Officer Closure

As a field officer, I want to update status and upload closure photo **so that** the report can be verified.

Acceptance Criteria:

- Officer app lists assignments, allows In Progress → Resolved
- Requires closure photo for Resolved, and appends history

US-6: Analytics

As a supervisor, I want to see counts, avg resolution time, and a heatmap **so that** I can plan resources.

Acceptance Criteria:

- Dashboard displays last 30d counts by category, avg time to resolve, open tickets per ward
- Heatmap indicates top 5 wards visually

US-7: Security

As a system admin, **I want** role-based access **so that** only authorized users can change statuses.

Acceptance Criteria:

- RBAC implemented: Reporter (public), FieldOfficer, Supervisor, Admin, Auditor
- Only Admin can change roles and SLAs

7. Data Model

Report Entity

```
{ id: "CM-YYYYMMDD-XXXX", // string title: string, description:
text, category: enum, photos: [URLs], voice_notes: [URLs],
location: { lat: number, lng: number, address: string, ward_id:
string }, status: enum, priority: enum, reporter: { id:
string/null, phone: string, email: string, anonymous: boolean },
assigned_to: string, // user id department_id: string, created_at:
timestamp, updated_at: timestamp, sla_due_at: timestamp, history:
[{ status: string, by: string, at: timestamp, comment: string,
media_url: string }] }
```

User Entity

```
{ id: string, name: string, role: enum, phone: string, email:
string, department_id: string, ward_ids: [string] }
```

Department Entity

```
{ id: string, name: string, contact: string, categories_handled:
[string] }
```

Ward Entity

```
{ id: string, name: string, boundary: geojson, parent_municipal:
string }
```

AuditLog Entity

```
{ id: string, action: string, actor_id: string, target_id: string,
timestamp: timestamp, details: object // immutable }
```

8. API Design

Public Endpoints

POST /api/v1/reports

Content-Type: multipart/form-data

Body: images, audio, json payload {category, description, lat, lng, address, reporter}

Response: {trackingId, reportId, createdAt}

GET /api/v1/reports/{trackingId}

Description: Public tracking (if private, require phone token)

Admin Endpoints (Auth Required)

GET /api/v1/admin/reports?

status=&category=&ward=&page=&limit=

Description: List reports with filtering and pagination

GET /api/v1/admin/reports/{id}

Description: Get detailed report information

PATCH /api/v1/admin/reports/{id}

Body: {status, assigned_to, comment, sla_due_at}

Description: Update report status and assignment

POST /api/v1/admin/reports/{id}/assign

Description: Assign report to field officer

Field Officer Endpoints

GET /api/v1/assignments?userId=

Description: Get assigned tasks for field officer

PATCH /api/v1/assignments/{id}/status

Description: Update assignment status (upload closure photo etc)

Analytics Endpoints

GET /api/v1/analytics/summary?from=&to=&ward=

Description: Get analytics summary for specified time period and ward

Authentication Endpoints

POST /api/v1/auth/login

Description: Admin/field officer login

POST /api/v1/auth/verify-phone

Description: OTP flow for reporters

Security Notes

- All admin endpoints protected with JWT (short TTL)
- Refresh tokens secure
- RBAC enforced server-side

9. System Workflows

Primary Report Processing Workflow

1 Citizen submits report (PWA)

Client compresses photo, sends to /reports endpoint

2 Server processes submission

Server stores media in object store (signed URL) and report in DB; calculates ward via geo lookup and category mapping; sets SLA due and initial status NEW

3 Auto-routing determines assignment

Auto-routing determines department and assigns (or places in unassigned queue)

4 Admin/dashboard receives report

Admin/dashboard receives report; if auto-assigned, field officer notified

5 Field officer processes task

Field officer accepts → In Progress → Resolves and uploads closure photo

6 Completion and monitoring

Reporter receives notification and may verify; supervisor can audit

7 Analytics and SLA monitoring

Analytics update (near real-time) and SLAs monitored; auto-escalation if SLA missed

10. Acceptance Criteria

(Testable)

- Submit report with image + GPS → returns tracking id within 3s
- Report appears on admin list within 5s
- Assigning a report notifies the field officer (push or SMS) within 5s
- Field officer can mark Resolved only after uploading closure photo
- Dashboard shows avg time to resolve computed correctly for resolved tickets
- Role-based endpoints deny unauthorized users (401/403)
- System rejects files >10MB or disallowed types

11. Priority Backlog

MVP (SIH / Grand Finale: 36–48 hours + polish)

- Core PWA report form (photo, GPS, category, submit)
- Basic admin dashboard (list, filter, detail, assign, status change)
- Rule-based routing (category→department + ward lookup CSV)
- Field officer minimal app (task list + closure photo)
- SMS mock-notifications + tracking page
- Basic analytics (counts + avg resolve time)
- DB + object store, simple auth for admin

30-day (Post-hackathon pilot)

- Multi-image, voice notes, offline PWA
- Configurable SLAs, escalate rules
- CSV export + schedule export
- Real SMS/Push integration (MSG91/Twilio + FCM)
- Improved UI/UX + multilingual strings

90-day (Scale & harden)

- Duplicate detection heuristics
- ML priority scoring prototype
- Integration with municipal asset register / work order
- Full security hardening, pen testing, and performance tuning
- Observability, monitoring, alerts and SRE runbooks

12. Security & Privacy Checklist

- Enforce TLS & HSTS for all endpoints
- Use signed URLs for media access, expire after time
- Encrypt PII at rest with KMS
- Implement OAuth2/OpenID Connect for admin (Keycloak recommended)
- Implement OTP verification for sensitive reporter flows
- Audit logs: store append-only (immutable), exportable
- Rate-limit public POST /reports (per IP + per phone)
- Daily vulnerability scans + weekly dependency checks

13. Testing Strategy

Unit Tests

Core business logic (routing, SLA calc, duplicate detection)

Integration Tests

APIs (mock object store)

End-to-End Tests

PWA → admin → field officer flows (Cypress or Playwright)

Load Testing

Media uploads and concurrent submission (k6 or Locust)

Security Testing

SAST + DAST + penetration test before pilot

14. Monitoring & KPIs

Adoption Metrics

- Reports/day
- Reports/100k citizens

Responsiveness Metrics

- % acknowledged within SLA
- % resolved within SLA

Quality Metrics

- Reopen rate
- Closure photo compliance
- Citizen satisfaction score

Operational Metrics

- API latency p95
- Error rate
- S3 storage growth per month

15. Developer-Ready Implementation Checklist

1. Bootstrap repo (monorepo or separate front/back) with templates
2. Implement DB schema (Postgres + PostGIS recommended) and object store bucket
3. Implement user auth (admin) and OTP phone flow (reporter)
4. Build PWA report form with client-side compression (resize to 1024px max) and EXIF handling
5. Implement POST /reports endpoint + media upload handling (signed URL or direct server)
6. Implement ward lookup microservice (lat/lng → ward id)
7. Implement routing engine (simple rules config in DB)
8. Implement admin dashboard (React) with list + detail + map (Leaflet)
9. Implement field officer mobile UI (React Native / PWA)
10. Hook notifications (SMS mock then live)
11. Implement analytics service (simple aggregator + charts)
12. Add logging, metrics, and error reporting
13. Run security checks, load tests
14. Deploy to staging and demo

16. SIH Demo Plan

Slide Content Structure

- **Problem Statement:** Jharkhand focus
- **PGMS Gap Summary:** 3 key bullets
- **CivicMitra Solution:** 3 key bullets
- **MVP Scope:** Core features demonstration
- **Demo Flow:** Step-by-step walkthrough

Live Demo Flow

1 Citizen Report Submission

Open PWA on phone → submit pothole with photo & pin → show returned tracking ID

2 Admin Processing

Admin dashboard receives new ticket (auto-route) → assign to officer

3 Field Officer Response

Field officer app receives task → marks In Progress → uploads closure photo → marks Resolved

4 Citizen Verification

Citizen tracking page shows status updated + closure photo

5 Analytics Dashboard

Show analytics page: counts & heatmap visualization

6 Security & SLA Highlights

Highlight security (RBAC) and SLA management features

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Empowering Citizens, Enabling Governance